
Function Notation & Evaluations

Practice Questions

Digital SAT Style | Mixed Difficulty

Total Questions: 50

Difficulty	Count
Easy	14
Medium	24
Hard	12
Grid-In Questions	4

Q1. DIFFICULTY: EASY

The function f is defined by $f(x) = 7x + 3$. What is the value of $f(5)$?

- A) 35
 - B) 38
 - C) 42
 - D) 53
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Q2. DIFFICULTY: EASY

The function g is defined by $g(x) = 4x - 9$. What is the value of $g(3)$?

- A) 3
 - B) 12
 - C) 21
 - D) -3
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Q3. DIFFICULTY: EASY

The function h is defined by $h(x) = -3x + 14$. What is the value of $h(-2)$?

- A) 8
 - B) 20
 - C) -20
 - D) 22
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Q4. DIFFICULTY: EASY

The function f is defined by $f(x) = 150x - 40$. What is the value of $f(x)$ when $x = 2$?

GRID-IN (Enter your answer in the box)

Q5. DIFFICULTY: EASY

If $f(x) = x^2 + 5$, what is the value of $f(4)$?

- A) 9
 - B) 13
 - C) 21
 - D) 69
-

Q6. DIFFICULTY: EASY

A function p is defined by $p(x) = 12 - 2x$. What is $p(7)$?

- A) -2
- B) 2
- C) -26

D) 26

Q7. DIFFICULTY: EASY

The function f is defined by $f(x) = x/4 + 6$. What is the value of $f(20)$?

- A) 11
 - B) 6.5
 - C) 26
 - D) 32
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Q8. DIFFICULTY: EASY

The function f is defined by $f(x) = 3(x + 4)$. What is the value of $f(6)$?

- A) 22
 - B) 30
 - C) 18
 - D) 42
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Q9. DIFFICULTY: MEDIUM

The function f is defined by $f(x) = 2x^2 - 3x + 1$. What is the value of $f(-1)$?

- A) 0
 - B) 4
 - C) 6
 - D) -4
-

Q10. DIFFICULTY: MEDIUM

If $g(x) = 5x - 8$, for what value of x is $g(x) = 17$?

- A) 5
 - B) 3
 - C) 9
 - D) 1.8
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Q11. DIFFICULTY: MEDIUM

Let $f(x) = 3x + 7$ and $g(x) = 2x - 1$. What is the value of $f(g(2))$?

- A) 16
 - B) 13
 - C) 19
 - D) 10
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Q12. DIFFICULTY: MEDIUM

The function f is defined by $f(x) = x^2 - 4x + 3$. What is $f(5) - f(1)$?

- A) 8
 - B) 0
 - C) 12
 - D) 4
-

Q13. DIFFICULTY: MEDIUM

The function f is defined by $f(x) = (x - 3)(x + 5)$. What is the value of $f(3)$?

- A) 0
 - B) 16
 - C) -16
 - D) 8
-

Q14. DIFFICULTY: MEDIUM

A function is defined by $f(x) = |2x - 10|$. What is the value of $f(3)$?

- A) -4
 - B) 4
 - C) 16
 - D) -16
-

Q15. DIFFICULTY: MEDIUM

The function f is defined by $f(x) = 3x^2 + 2$. What is the value of $f(a + 1)$ in terms of a ?

- A) $3a^2 + 6a + 5$
 - B) $3a^2 + 3$
 - C) $3a^2 + 6a + 3$
 - D) $3a^2 + 2a + 5$
-

Q16. DIFFICULTY: MEDIUM

If $f(x) = 6x - 4$, what is the value of $f(2a)$?

- A) $12a - 4$
 - B) $6a - 4$
 - C) $12a - 8$
 - D) $2(6a - 4)$
-

Q17. DIFFICULTY: MEDIUM

The function h is defined by $h(t) = -16t^2 + 80t + 4$. What is $h(2)$?

- A) 100
 - B) 96
 - C) 164
 - D) 228
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Q18. DIFFICULTY: MEDIUM

A linear function f is defined such that $f(0) = 8$ and $f(4) = 20$. What is $f(6)$?

- A) 24
 - B) 26
 - C) 28
 - D) 32
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Q19. DIFFICULTY: MEDIUM

If $f(x) = 4x + 1$, what is $f(x + 3) - f(x)$?

- A) 3
 - B) 7
 - C) 12
 - D) 4
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Q20. DIFFICULTY: MEDIUM

If $f(x) = x^2 - 9$ and $g(x) = x + 3$, what is $f(x)/g(x)$ for $x \neq -3$?

- A) $x - 3$
 - B) $x + 3$
 - C) $x^2 - 3$
 - D) $x - 9$
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Q21. DIFFICULTY: MEDIUM

The function f is defined by $f(x) = 2x + 5$. If $f(k) = 19$, what is the value of k ?

GRID-IN (Enter your answer in the box)

Q22. DIFFICULTY: MEDIUM

The function g is defined by $g(x) = -x^2 + 6x$. What is the maximum value of $g(x)$?

- A) 6
 - B) 9
 - C) 3
 - D) 12
-

Q23. DIFFICULTY: MEDIUM

If $f(x) = 3x - 2$ and $g(x) = x^2$, what is $g(f(1))$?

- A) 1
 - B) 3
 - C) 7
 - D) 9
-

Q24. DIFFICULTY: HARD

The function f is defined by $f(x) = ax^2 + bx + c$, where a , b , and c are constants. If $f(0) = 5$, $f(1) = 5$, and $f(-1) = 3$, what is the value of a ?

- A) -1
 - B) 1
 - C) 2
 - D) -2
-

Q25. DIFFICULTY: HARD

If $f(x) = 2x + 3$, what is $f(f(x))$?

- A) $4x + 6$
 - B) $4x + 9$
 - C) $2x + 9$
 - D) $4x^2 + 12x + 9$
-

Q26. DIFFICULTY: HARD

If $f(x) = 3x - 1$ and $f(g(x)) = 6x + 5$, what is $g(x)$?

- A) $2x + 2$
 - B) $2x + 1$
 - C) $2x + 5$
 - D) $3x + 2$
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Q27. DIFFICULTY: EASY

The function f is defined by $f(x) = 85x - 15$. What is $f(x)$ when $x = 1$?

GRID-IN (Enter your answer in the box)

Q28. DIFFICULTY: EASY

If $f(x) = -5x + 20$, what is $f(0)$?

- A) 20
- B) 0

- C) -5
 - D) 25
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Q29. DIFFICULTY: EASY

The function g is defined by $g(n) = n/3 - 4$. What is $g(15)$?

- A) 1
 - B) 5
 - C) 11
 - D) -1
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Q30. DIFFICULTY: MEDIUM

If $f(x) = x^2 + 2x$ and $g(x) = x - 4$, what is $f(g(5))$?

- A) 3
 - B) 35
 - C) 8
 - D) 0
-

Q31. DIFFICULTY: MEDIUM

The function f is defined by $f(x) = 3x^2 - 12$. For what positive value of x does $f(x) = 0$?

- A) 2
 - B) 4
 - C) 6
 - D) 12
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Q32. DIFFICULTY: HARD

The function f is defined by $f(x) = (2x - 1)/(x + 3)$. What is $f(5)$?

- A) $9/8$
 - B) $11/8$
 - C) $8/9$
 - D) $3/2$
-

Q33. DIFFICULTY: HARD

If $f(x) = x^3 - 2x^2 + x - 3$, what is $f(2)$?

- A) -1
 - B) 1
 - C) -3
 - D) 3
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Q34. DIFFICULTY: EASY

If $f(x) = 10 - x$, what is $f(10)$?

- A) 0
 - B) 10
 - C) 20
 - D) -10
-

Q35. DIFFICULTY: MEDIUM

A function f is defined as $f(x) = 4x - 7$. Which of the following is equivalent to $2 \cdot f(x) + 3$?

- A) $8x - 11$
 - B) $8x - 14$
 - C) $8x - 4$
 - D) $8x + 3$
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Q36. DIFFICULTY: MEDIUM

The function f is defined by $f(x) = 5x + 2$. If $f(a) = f(b) + 10$, which of the following expresses a in terms of b ?

- A) $a = b + 2$
 - B) $a = b + 10$
 - C) $a = b + 5$
 - D) $a = 5b + 12$
-

Q37. DIFFICULTY: HARD

Let $f(x) = x^2 - 6x + 8$. For how many values of x does $f(x) = f(x + 4)$?

- A) 0
 - B) 1
 - C) 2
 - D) Infinitely many
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Q38. DIFFICULTY: EASY

The function f is defined by $f(x) = 200x + 50$. The function f gives the total cost, in dollars, for renting x hours of studio time. What is the best interpretation of $f(3) = 650$?

- A) The cost of renting 3 hours of studio time is \$650.
 - B) The cost of renting 650 hours of studio time is \$3.
 - C) For every 3 hours, the cost increases by \$650.
 - D) The studio charges \$3 for 650 minutes.
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Q39. DIFFICULTY: MEDIUM

If $f(x) = x^2 - 8$ and $g(x) = 2x$, for what positive value of x does $f(x) = g(x)$?

GRID-IN (Enter your answer in the box)

Q40. DIFFICULTY: HARD

The function f is defined by $f(x) = (x + 2)^2 - 9$. Which of the following is an equivalent form that displays the zeros of f as constants?

- A) $f(x) = (x + 5)(x - 1)$
 - B) $f(x) = (x - 5)(x + 1)$
 - C) $f(x) = (x + 3)(x - 3)$
 - D) $f(x) = x^2 + 4x - 5$
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Q41. DIFFICULTY: MEDIUM

If $f(x) = 3x - 5$, what value of x makes $f(x)$ equal to x ?

- A) $5/2$
 - B) 5
 - C) $5/4$
 - D) $-5/2$
-

Q42. DIFFICULTY: HARD

The function f is defined for all real numbers. The table below shows selected values of f .

x	-2	-1	0	1	2	7
$f(x)$	-2	-1	0	1	4	7

Based on the table, which of the following could define f ?

- A) $f(x) = x^2 + 3$
 - B) $f(x) = 2x^2 + 3$
 - C) $f(x) = x^2 - 3$
 - D) $f(x) = 3x^2 + 3$
-

Q43. DIFFICULTY: MEDIUM

If $f(x) = x^2$ and $g(x) = f(x - 3) + 2$, which of the following defines g ?

- A) $g(x) = (x - 3)^2 + 2$
 - B) $g(x) = x^2 - 1$
 - C) $g(x) = (x + 3)^2 + 2$
 - D) $g(x) = x^2 + 5$
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Q44. DIFFICULTY: HARD

If $f(x) = ax + b$, $f(2) = 11$, and $f(5) = 23$, what is the value of $f(0)$?

- A) 3

- B) 5
 - C) 1
 - D) 7
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Q45. DIFFICULTY: EASY

The function f is defined by $f(x) = 8$. What is $f(100)$?

- A) 100
 - B) 8
 - C) 800
 - D) 108
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Q46. DIFFICULTY: MEDIUM

If $f(x) = 2x$, which of the following is equal to $f(3) + f(5)$?

- A) $f(8)$
 - B) $f(6)$
 - C) $f(15)$
 - D) $f(7)$
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Q47. DIFFICULTY: HARD

The function f is defined by $f(x) = (x - 1)^2 + k$, where k is a constant. If $f(4) = 15$, what is the value of k ?

- A) 6
 - B) 9
 - C) 3
 - D) 24
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Q48. DIFFICULTY: HARD

The function f is defined by $f(x) = 2(3)^x$. What is the value of $f(4) - f(3)$?

- A) 2
 - B) 54
 - C) 108
 - D) 162
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Q49. DIFFICULTY: MEDIUM

If $f(x) = -2x + 10$, what is the value of x when $f(x) = f(-x)$?

- A) 0
- B) 5

- C) -5
 - D) 10
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Q50. DIFFICULTY: HARD

The functions f and g are defined as $f(x) = x^2 + 1$ and $g(x) = 2x - 3$. If $h(x) = f(g(x))$, what is $h(2)$?

- A) 2
 - B) 5
 - C) 10
 - D) 4
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